Ch 02: Introduction to Turtle Graphics

Application Program Development

Derek Harter

Department of Computer Science
East Texas A&M University

Summer 2026

EAST TEXAS A&M

Introduction to Turtle Graphics

Why Turtle Graphics?

Turtle graphics is an interesting and easy way to learn basic programming concepts. The Python turtle graphics system simulates a "turtle" that obeys commands to draw simple graphics.

EAST TEXAS A&M

Introduction to Turtle Graphics (1 of 3)

```
import turtle
turtle.forward(200)
```

- `showturtle()` turn turtle on, make lines drawn visible
- `forward(n)` move turtle distance `n`
- `right(angle)` `left(angle)` turn turtle left or righ by relative angle
- `setheading(angle)` face turtle in towards absolute heading
- `heading()` return current heading of turtle
- `penup()` `pendown()` raise or lower turtle pen, causes line to be drawn or not
- `circle(radius)` draw a circle of given radius
- `dot()` draw a dot at current turtle location
- `pencsize(width)` change width of pencil drawing line
- `pencolor('color')` change pen color
- `bgcolor('color')` change drawing canvas background color
- `reset()` erases all drawings currently in windows, reset color to `black`, reset turtle origin



EAST TEXAS A&M

Introduction to Turtle Graphics (2 of 3)

- `clear()` simply erases all drawings, doesn't change turtle position or color
- `clearscreen()` erases all drawings, reset graphics windows
- `setup(width, height)` specify size of graphics windows
- `goto(x,y)` move turtle to coordinate position
- `x,y = pos()` get the turtle's current coordinate position
- `xcor()` `ycor()` get the x and y coordinate respectively
- `speed(speed)` change turtle speed
- `hideturtle()` hide the turtle
- `write('text')` display text in the graphics windows
- `begin_fill()` `end_fill()` `fill_color('color')` fill in shaps when drawn
- `value = numinput('title', 'prompt')` pop up dialog to enter an integer value
- `done()` keep window open if running in script mode



Introduction to Turtle Graphics (3 of 3)

```
(base) student@appdev:/workspaces/csci233-appdev/code/ch02$ python3 orion.py
```

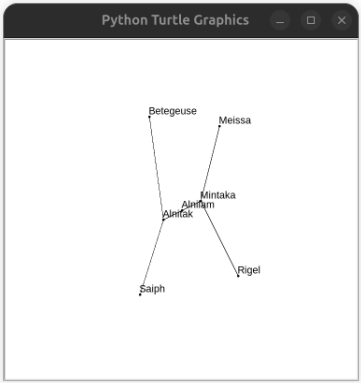


Figure 1: Output of the ‘orion.py‘ program

EAST TEXAS A&M

Summary

- Turtle graphics is an interesting way to learn basic programming concepts
- You can graphically results of using the `turtle` graphics module
- Commands to move, rotate, draw, color and add text to graphical output

EAST TEXAS A&M

References



EAST TEXAS A&M